cmp JobPortal architecture

«device»
User's computer

Job seeker

uses

«executionEnvironment»
App window
{native window, app mode or browser tab}

«component»
Dashboard
index.html, style.css, js/
pages: Home, Leads, Tracker, Analysis, Settings

opens

REST API
(JSON, /api/*)

Event stream
(SSE, /api/events)

«executionEnvironment»
JobPortal server
{Python, FastAPI, 127.0.0.1:8788}

«component»
app
cli.py, desktop.py, doctor.py

starts

«component»
web
server.py, common.py, routes/

sign-in

domain

«component»
search
pipeline.py, plan.py, packages.py,
sources.py, collectors/

«component»
insights
analysis.py

«component»
documents
extract.py, render.py,
resumecheck.py, exports.py,
revise.py

«component»
candidate
onboarding.py, profile.py,
documents.py, library.py

«component»
ai
client.py, signin.py,
prompts.py

«component»
core
config.py, settings.py, db.py, bus.py,
background.py, osutil.py, textutil.py,
levels.py, genres.py

«external system»
Social and community feeds
Reddit, Mastodon, Bluesky,
Hacker News, RSS feeds

HTTPS

«external system»
Job boards and job APIs
LinkedIn, company job boards,
USAJobs, The Muse, Adzuna,
Google Jobs, remote job boards

HTTPS

«external system»
OpenAI
Sign in with ChatGPT
(OAuth 2.0 with PKCE),
Responses API (JSON schemas)

HTML to PDF

SQL

«executable»
Chrome Headless Shell
renders resumes and cover
letters to PDF

«database»
jobportal.db
SQLite: kv, leads, events, runs, run_log,
candidates, ai_calls, resume_library

Every package uses core:
settings, the database,
the event bus and
background work.

Data flow
1. Setup: the Dashboard sends the resume, documents and answers to web. candidate reads them (documents extracts the text) and ai turns them into the profile, which core saves in jobportal.db.
2. Search: search collects postings from the job sources, filters them by the user's targets, asks ai to score the rest and write application packages, and saves the leads.
3. Live updates: core's event bus streams progress and new leads to the Dashboard over the event stream.
4. Tracking: status changes go back through the REST API to jobportal.db.